

YI LI

🌐 <https://yili.vision> ✉ yili.matrix@gmail.com

🎓 Scholar 🔗 LinkedIn

RESEARCH AND ENGINEERING FOCUS

Pretraining data for frontier LLMs at trillion-token scale: web-scale processing, quality filtering, corpus coverage, interleaved web data, token efficiency, and training stability.

WORK EXPERIENCE

xAI, Pretraining

June 2025 – May 2026

Member of Technical Staff

- Owned pretraining web data pipelines; processed trillions of web tokens for Grok pretraining.
- Built a web-interleaved data pipeline producing $\sim 1\text{T}$ tokens for pretraining and image generation.
- Designed and shipped text and image quality filters and corpus coverage checks to improve data quality and token efficiency.
- Built online sequence-outlier masking to reduce grad-norm spikes and stabilize pretraining.
- Curated hundreds of billions of high-quality tokens from X, Wikipedia, and Wikimedia.

NVIDIA, AI Robotics Research Lab

Jan. 2024 – June 2025

Research Intern

supervised by Dr. Ankit Goyal

- Led HAMSTER, a hierarchical Vision-Language-Action (VLA) model for open-world robot manipulation that enabled visual, semantic, and geometric generalization across domain gaps. Accepted to ICLR 2025.

Microsoft Research Asia, Visual Computing Group

Nov. 2015 – Jun. 2017

Research Intern

supervised by Dr. Jifeng Dai

- Co-developed Deformable Convolutional Networks, introducing deformable convolution and RoI pooling for geometric variation in images. Accepted to ICCV 2017 (oral).
- Co-developed FCIS, an end-to-end instance segmentation framework accepted to CVPR 2017 (spotlight); won 1st Prize in the MS COCO 2016 Segmentation Challenge.
- Co-developed R-FCN, a fast and accurate region-based fully convolutional object detector accepted to NeurIPS 2016.

EDUCATION

University of Washington

2018 – 2025

Ph.D., advised by Prof. Dieter Fox

Computer Science and Engineering

Tsinghua University

2014 – 2017

Master of Science, with highest distinction

Automation

Tsinghua University

2010 – 2014

Bachelor of Engineering (National Scholarship)

Automation

SELECTED PUBLICATIONS

(* indicates co-first author, full list at Google Scholar)

Yi Li*, Yuquan Deng*, Jesse Zhang*, Joel Jang, Marius Memmel, Caelan Garrett, Fabio Ramos, Dieter Fox, Anqi Li, Abhishek Gupta, Ankit Goyal

HAMSTER: Hierarchical Action Models for Open-World Robot Manipulation

International Conference on Learning Representations (ICLR), 2025. [Project website](#)

Invited presentations at OpenAI, Google DeepMind, and ByteDance Seed

Yi Li, Gu Wang, Xiangyang Ji, Yu Xiang, Dieter Fox

DeepIM: Deep Iterative Matching for Object Pose Estimation

In *International Journal of Computer Vision (IJCV)*, 2020, [arXiv](#)

In *European Conference on Computer Vision (ECCV)*, 2018 (oral)

Selected as one of the top 12 papers in ECCV 2018

Jifeng Dai*, Haozhi Qi*, Yuwen Xiong*, Yi Li*, Guodong Zhang*, Han Hu, Yichen Wei

Deformable Convolutional Networks

In *International Conference on Computer Vision (ICCV)*, 2017 (oral). [arXiv](#)

Ranked 6th among the most influential papers in ICCV 2017

Yi Li*, Haozhi Qi*, Jifeng Dai, Xiangyang Ji, Yichen Wei

Fully Convolutional Instance-aware Semantic Segmentation

In *Computer Vision and Pattern Recognition (CVPR)*, 2017 (spotlight). [arXiv](#)

1st Prize in MS COCO 2016 Segmentation Challenge

Jifeng Dai, Yi Li, Kaiming He, Jian Sun

R-FCN: Object Detection via Region-Based Fully Convolutional Networks

In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016. [arXiv](#)

Ranked 2nd among the most influential papers in NeurIPS 2016

PROFESSIONAL SERVICE AND HONORS

Served as a reviewer for top-tier conferences and journals, including ICLR, CVPR, ICCV, ECCV, RA-L, AAI, ICRA, and IROS, since 2017.

Outstanding 2017 Master's Thesis, awarded by the Chinese Institute of Electronics (top 10 in China)

1st Prize in MS COCO 2016 Object Detection Challenge

Outstanding 2016 Intern at MSRA

2017 Summa Cum Laude in Beijing (top 2% at Tsinghua)

2013 National Scholarship (top 0.5% nationwide)